

Adopted Levels: not observed

$S(n)=24210$ syst; $S(p)=-60$ syst; $Q(\alpha)=-9610$ syst 2021Wa16

$\Delta S(n)=500$, $\Delta S(p)=360$, $\Delta Q(\alpha)=350$ (syst, 2021Wa16).

$S(2p)=-2510$ 300, $Q(\epsilon)=16110$ 360, $Q(\epsilon p)=16990$ 300 (syst, 2021Wa16).

Isotope discovery: none.

$S(2p)$ is negative and $S(p)$ may be consistent with zero. ^{34}Ca is a candidate for ground-state two-proton emitter.

2026Li03: $S(p)=401$ and $S(2p)=-1987$ from an average of Finite-Range Droplet Model (FRDM12), Koura-Tachibana-Uno-Yamada (KTUY) model, Weizsacker-Skyrme (WS4) model, and Duflo-Zuker (DZ31) model combined with the radial basis function mirror symmetry (RBFms) approach.

2024Li59: $S(2p)=-1044$, $T_{1/2}=177.8$ ns. The $Q(2p)$ decay energy was determined using an empirical linear relationship. The $T_{1/2}$ was calculated by inputting the refined decay energy into the new Geiger-Nuttall formula.

2024Ji17: $T_{1/2}=8.1$ fs from an extension of the Bayrak model with and without the deformation term. Used $Q(2p)=2510$ from 2021Wa16.

2023Me08: $T_{1/2}=64.6$ ps. The WKB approximation is applied to a Yukawa-plus-exponential model potential. Used $Q(2p)=1470$ from 2017Wa10 and a spectroscopic factor derived from a specific cubic-polynomial relation.

2022Zo01: $S(p)=259$ 96, $S(2p)=-2132$ 151, mass excess=14497 98. Mass relations of mirror nuclei.

2022Sa41: $T_{1/2}=2.2$ fs from the Coulomb and proximity potential model for deformed nuclei (CPPMDN) incorporating ground-state deformation (β_2) and orientation effect. Used $Q(2p)=2510$ from 2021Wa16.

2022Ro15: $T_{1/2}=0.35$ fs from the generalized liquid drop model (GLDM). $T_{1/2}=3.0$ as from a semi-empirical analytic expression based on a modified version of the Geiger-Nuttall law. Both calculations used $Q(2p)=2510$ from 2021Wa16.

2022Pa24: $T_{1/2}=0.31$ ns from a two-potential approach; the nuclear potential is obtained using the Skyrme-Hartree-Fock approach and the Skyrme effective interaction of SLy8. Used $Q(2p)=1470$ from 2017Wa10.

2021Xi06: $T_{1/2}=3.5$ fs from the Unified Fission Model (UFM), $T_{1/2}=1.7$ fs from the Generalized Liquid Drop Model (GLDM), $T_{1/2}=27.5$ fs from the Effective Liquid Drop Model (ELDM). These models share a similar physical mechanism by treating the decay as a ^2He cluster preformed on the nuclear surface that subsequently penetrates the Coulomb barrier. All used $Q(2p)=2510$ from 2021Wa16.

2021Li27: $T_{1/2}=79.4$ ps from the Gamow-like model. Used $Q(2p)=1470$ from 2017Wa10.

2021Li16: $T_{1/2}=117$ ps from a two-parameter empirical formula based on the Geiger-Nuttall law. Used $Q(2p)=1470$ from 2017Wa10.

2020Ma35: $S(p)=203$ 151, $S(2p)=-2152$ 118, $T_{1/2}=3.6$ ps $+51-20$. Mass relations of mirror nuclei with local correlations and a four-parameter empirical formula based on the Effective Liquid Drop Model (ELDM).

2020Cu01: $T_{1/2}=19.5$ ps from a generalized liquid drop model (GLDM). Used $Q(2p)=1470$ from 2017Wa10.

2019Sr02: $T_{1/2}=2.2$ ns. A four-parameter empirical formula was fitted using the theoretical results of the Effective Liquid Drop Model (ELDM). Used $Q(2p)=1470$ from 2017Wa10.

Other decay half-life calculations: 2017Sa70, 2004Pf02, 2003Gr04, 2005Pf01, 2003Gr24, 2001Gr29.

Other structure calculations: 2005Wa15, 2005Zh28, 2002Zh09, 2001Yo12, 1998Co30, 1997Co19, 1997Pa38, 1996Sh20, 1995La03.

Giant-resonance calculations: 2021Li26, 2001Sa57, 1997Ha57, 2003Ma71, 2001Sa77, 1999Sa22, 1998Ha09, 1998Ha39, 1998Sa28, 1997Ha55.

A two-hour search conducted on March 27, 2012, during the 2015Su01 experiment at the Radioactive Ion Beam Line in Lanzhou (RIBLL), did not observe any events at ~ 200 ns that could be attributed to ^{34}Ca in the ΔE -ToF spectrum.

 ^{34}Ca Levels

<u>E(level)</u>	<u>Jπ</u>	<u>T$_{1/2}$</u>	<u>Comments</u>
0?	0 ⁺	<35 ns	%p=?; %2p=? $T_{1/2}$: upper limit estimated from expected cross sections of a ^{40}Ca beam fragmentation on a Ni target at LISE, GANIL (1996PoZZ). The authors of 1986La17 provided additional data from their experiment in a private communication 1996PoZZ.